

INX Future Inc. Employee Performance Analysis

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Table of Contents

Purpose of the Analysis	3
Data Understanding	3
Data Preparation	4
1.1 Removal of Employee Number	4
1.2 Separation of Features and Target	4
1.3 Categorical Variable Encoding	4
1.4 Training and Testing Data	5
Exploratory Data Analysis Approach	5
Feature Selection and Engineering	6
Machine Learning Algorithm Selection	6
Model Training Approach	7
Model Evaluation	7
Model Selection Decision	8
Feature Importance Analysis	8
Prediction Approach	9
Visualization Approach	9
Tools and Techniques Used	10
Analysis Conclusion	10

Purpose of the Analysis

The purpose of this analysis was to examine the employee performance dataset and develop a machine learning approach for predicting employee performance ratings.

The analysis focused on understanding the structure and quality of the data, identifying useful relationships between employee characteristics and performance, preparing the data for machine learning, comparing different classification algorithms, and selecting the most suitable model for prediction.

Data Understanding

The dataset contains employee information such as demographic characteristics, education, department, job role, job satisfaction, work experience, salary hike percentage, work-life balance, and other workplace-related factors.

The target variable selected for prediction was: Performance Rating

The remaining relevant employee attributes were treated as input features.

The first stage of the analysis involved examining:

- Number of observations and variables
- Data types
- Summary statistics
- Missing values
- Duplicate records
- Distribution of the target variable

The dataset contained 1,200 employee records and 28 variables.

The data quality checks showed that there were no missing values and no duplicate records. Therefore, no missing-value imputation or duplicate removal was required.

Data Preparation

Several preprocessing steps were carried out before model training.

1.1 Removal of Employee Number

EmpNumber was removed from the modelling dataset because it is an employee identification number rather than a meaningful characteristic that should influence performance prediction. Including an identification number could introduce irrelevant information into the model.

1.2 Separation of Features and Target

The dataset was divided into:

- Input features (X): employee characteristics
- Target variable (y): Performance Rating

This separation allowed the machine learning algorithms to learn the relationship between employee characteristics and performance ratings.

1.3 Categorical Variable Encoding

The dataset contained categorical variables such as:

- Gender
- Education background
- Department
- Job role
- Business travel frequency
- Marital status
- Overtime
- Attrition

Machine learning algorithms require numerical input, so categorical variables were converted into numerical representations using one-hot encoding. `drop_first=True` was used to avoid unnecessary duplicate information between categories.

1.4 Training and Testing Data

The dataset was divided into training and testing. The training set was used to train the models while the testing set was used to evaluate how well the models performed on previously unseen data.

Exploratory Data Analysis Approach

Exploratory Data Analysis was performed before modelling to understand the patterns and relationships within the dataset. Different visualizations were selected depending on the type of variables being analyzed.

The analysis included:

- Overall performance rating distribution
- Department and performance rating
- Gender and performance rating
- Education level and performance
- Job role and performance
- Years at the company and performance
- Job satisfaction and performance
- Work-life balance and performance
- Salary hike percentage and performance
- Environment satisfaction and performance

The purpose of the EDA was not only to create visualizations but also to determine which employee characteristics appeared useful for further modelling.

Feature Selection and Engineering

The original dataset contained a large number of employee attributes. Rather than manually selecting only a few variables before modelling, the available relevant employee features were initially retained so that the machine learning algorithms could identify useful patterns.

The main feature preparation techniques were:

- Removing the employee identification number
- Encoding categorical variables
- Retaining numerical variables
- Maintaining the original target variable
- Scaling features where required by the algorithm

Machine Learning Algorithm Selection

Since Performance Rating represents categories/classes the project was treated as a classification problem.

Several classification algorithms were considered:

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. Gradient Boosting

The purpose of testing multiple algorithms was to determine which approach could provide the most reliable predictions rather than assuming that one algorithm would automatically perform best.

Model Training Approach

The models were trained using the prepared training dataset.

For Logistic Regression, feature scaling was applied using StandardScaler because Logistic Regression can benefit from features being placed on comparable numerical scales.

The tree-based algorithms: Decision Tree, Random Forest and Gradient Boosting were trained using the encoded but unscaled features because tree-based models do not require standardization in the same way. This approach ensured that preprocessing was appropriate for each algorithm.

Model Evaluation

The models were evaluated using several classification metrics:

- Accuracy- Measures the proportion of predictions that were correct.
- Precision- Measures how many of the observations predicted as a particular class were actually members of that class.
- Recall- Measures how successfully the model identified observations belonging to each class.
- F1-score- Provides a balance between precision and recall.

Using multiple evaluation metrics provided a more complete assessment of model performance than relying only on accuracy.

The results were compared across the four algorithms.

Model	Accuracy
Logistic Regression	82.50%
Decision Tree	90.83%
Random Forest	91.67%
Gradient Boosting	93.75%

Model Selection Decision

Gradient Boosting was selected as the final model. The decision was based on its performance across the evaluation metrics. It achieved the highest accuracy among the tested models, with an accuracy of 93.75%. It also produced the strongest overall combination of precision, recall, and F1-score. Therefore, Gradient Boosting was considered the most suitable model among the tested algorithms for the employee performance prediction task.

Feature Importance Analysis

After selecting Gradient Boosting, feature importance was examined to understand which variables contributed most to the model's predictions. The feature importance results were used to identify the three most influential features.

The three highest-ranked features were:

1. Employee Environment Satisfaction
2. Employee Last Salary Hike Percentage
3. Years Since Last Promotion

Feature importance indicates how useful a variable was to the model's predictions. It should not automatically be interpreted as proof that the variable directly causes employee performance to increase or decrease. This distinction is important when interpreting machine learning results.

Prediction Approach

After training and selecting the final model, the model was saved so that it could be reused for prediction. The prediction process involves:

1. Providing employee information.
2. Applying the same preprocessing used during training.
3. Aligning the input features with the model's expected feature structure.
4. Passing the processed employee information to the Gradient Boosting model.
5. Producing a predicted employee performance rating.

This approach allows the trained model to be used on individual employee information rather than only on the original dataset.

Visualization Approach

Visualizations were included throughout the project to make the analysis easier to interpret. The visualization notebook contains charts showing:

- Performance rating distribution
- Department performance
- Model performance comparison
- Important features identified by the final model

The visualizations support the numerical analysis and make patterns easier to communicate to non-technical stakeholders.

Tools and Techniques Used

The following tools and techniques were used during the project:

- Programming Language- Python
- Data Processing- Pandas, NumPy
- Data Visualization- Matplotlib , Seaborn
- Machine Learning- Scikit-learn
- Model Persistence- Joblib
- Development Environment- Jupyter Notebook / Google Colab

The main techniques included data cleaning, exploratory analysis, categorical encoding, feature preprocessing, classification, model evaluation, feature importance analysis and prediction.

Analysis Conclusion

The analysis established a complete workflow for employee performance prediction. The dataset was inspected and prepared, exploratory analysis was performed, several classification algorithms were evaluated and the models were compared using multiple performance metrics. Gradient Boosting was selected as the final modelling approach because it provided the strongest predictive performance among the algorithms tested. The feature importance analysis was then used to understand which employee related variables were most influential to the model's predictions.